

Less to Lose:

How shorter and more flexible degrees reshaped university choice and earnings

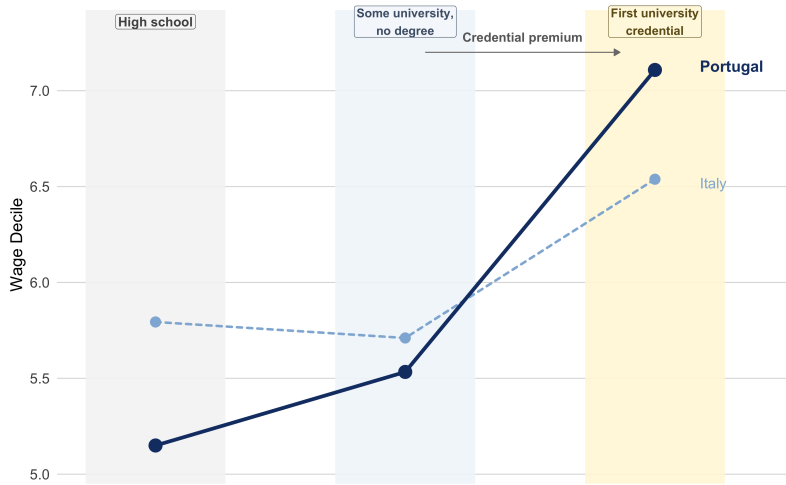
Emma Scandolo

University College London

September 17, 2026

University is risky because the payoff is lumpy

The wage premium appears concentrated at the first credential



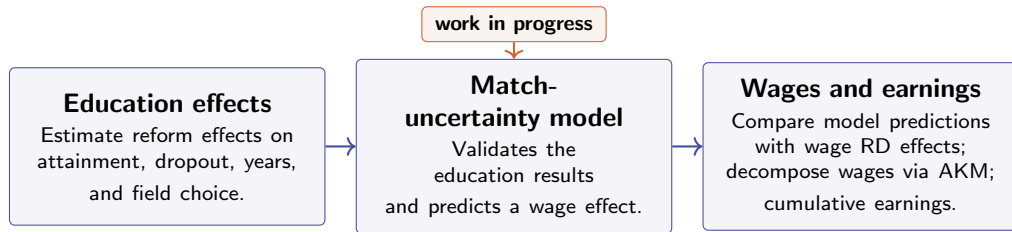
Source: PIAAC - Survey of Adult Skills, EU sample.

Notes: Wage deciles are age-adjusted. Sample restricted to employed adults.

Lowering the downside risk of enrollment

- ▶ Students enroll unsure whether university is a good fit
 - ▷ Downside risk deters marginal students from enrolling
- ▶ An earlier or new intermediate credential can change the bet
 - ▷ Reaching a credential sooner → ↓ forgone earnings and dropout risk
 - ▷ **Less to lose** from trying
- ▶ **Bologna Process** (1999): a large-scale natural experiment
 - ▷ 40+ countries: **4-5 year first degree** → **3 year** (+ master)
 - ▷ ~ 38 million students affected
- ▶ How did this affect *university choice, earnings, and firm sorting*?

This paper



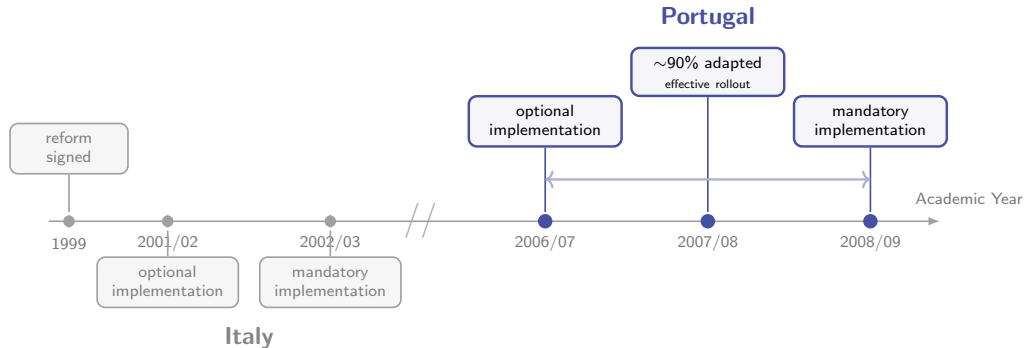
- **Portugal & Italy:** fast rollout → cohort RD
 - ▷ **Portugal:** linked employer-employee + survey data
 - ▷ **Italy:** survey data for external validity

Contribution

- ▶ **Bologna Reform** Cardoso et al. (2008); Cappellari and Lucifora (2009); Bondonio and Berton (2018); Avdeev (2025)
 - ▷ Prior work: enrollment, dropout.
 - ▷ **This paper:** completion, dropout, field choice, earnings, and firms.
- ▶ **Lower-risk tertiary access** Mountjoy (2022); Jepsen et al. (2014); Stevens et al. (2019)
 - ▷ Community colleges: lower-barrier access through **alternative institutions**.
 - ▷ **This paper:** credential design **inside universities**, with individuals followed over time into earnings and firms.
- ▶ **Education as experimentation** Manski (1989); Stinebrickner and Stinebrickner (2012); Trachter (2015)
 - ▷ Learning models: students learn about match quality after enrolling.
 - ▷ **This paper:** uses a policy-induced change in the choice set to validate the learning mechanism.

Background

Reform background: implementation timing



Sources:

Portugal: Cardoso et al. (2008), Portuguese FHEQ (2010); Italy: Cappellari and Lucifora (2009), Bondonio and Berton (2018).

Reform background: degree margins by subject

	Portugal		Italy		
	pre	post	pre	post	
STEM	5	3+2	5	3+2	new 3-year credential
Economics, social sciences, humanities, arts	4+2	3+2	4	3+2	first credential one year earlier
Nursing	4	4	3	3+2	Portugal: no change Italy: upgrade option
Medicine and law	unchanged		unchanged		no major change

Data and RD design

Data

1. Quadros de Pessoal (QP): 2000–2023

- ▷ Linked employer–employee dataset covering the private sector
- ▷ Variables: year of birth, gender, education level and field, wages
- ! *Conditional* on private-sector employment

2. Labour Force Surveys (LFS): 2000–2023

- ▷ Repeated cross-section; self-reported education
- ! *Unconditional* on employment type

3. Education and Training Survey (IEFA): 2007–2022

- ▷ Dropout rates
- ! *Small sample size*

Data

1. Quadros de Pessoal (QP): 2000–2023

- ▷ Linked employer–employee dataset covering the private sector
- ▷ Variables: year of birth, gender, education level and field, wages
- ! *Conditional* on private-sector employment

2. Labour Force Surveys (LFS): 2000–2023

Italy external validity
LFS: 2009–2023

- ▷ Repeated cross-section; self-reported education
- ! *Unconditional* on employment type

3. Education and Training Survey (IEFA): 2007–2022

- ▷ Dropout rates
- ! *Small sample size*

Fuzzy RD design

- ▶ Running variable: $Z_i = c_i - 2007$, $c_i = \text{year of birth} + 18$
- ▶ Treatment dummy: $D_i = \mathbf{1}[Z_i \geq 0]$ (cohort reached 18 after reform)
- ▶ Reform effect α_1 :

$$Y_i = \alpha_0 + \alpha_1 D_i + \alpha_2 Z_i + \alpha_3 Z_i \times D_i + \alpha_4 \mathbf{X}_i + \varepsilon_i$$

where $\mathbf{X}_i$ contains age polynomials

- ▶ Bandwidth: ± 6 birth cohorts around the cutoff
- ▶ Donut setting: exclude 2006 cohort

Descriptives

No choice delay

McCrary test

Fuzzy RD design

Italy: 2002

Italy: +19

- ▶ Running variable: $Z_i = c_i - 2007$, $c_i = \text{year of birth} + 18$
- ▶ Treatment dummy: $D_i = \mathbf{1}[Z_i \geq 0]$ (cohort reached 18 after reform)
- ▶ Reform effect α_1 :

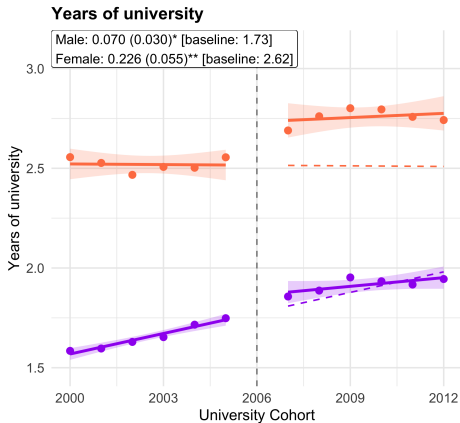
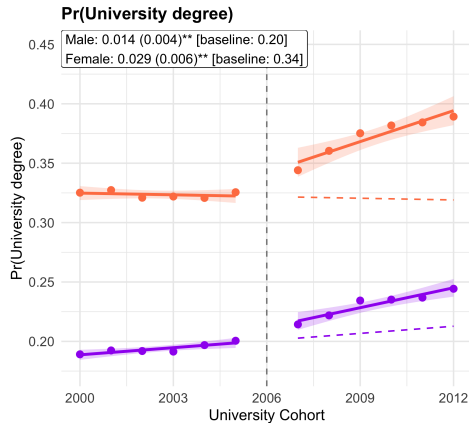
$$Y_i = \alpha_0 + \alpha_1 D_i + \alpha_2 Z_i + \alpha_3 Z_i \times D_i + \alpha_4 \mathbf{X}_i + \varepsilon_i$$

where $\mathbf{X}_i$ contains age polynomials

- ▶ Bandwidth: ± 6 birth cohorts around the cutoff
- ▶ Donut setting: exclude 2006 cohort

Education

↑ university completion (8–9%) and ↑ years of schooling



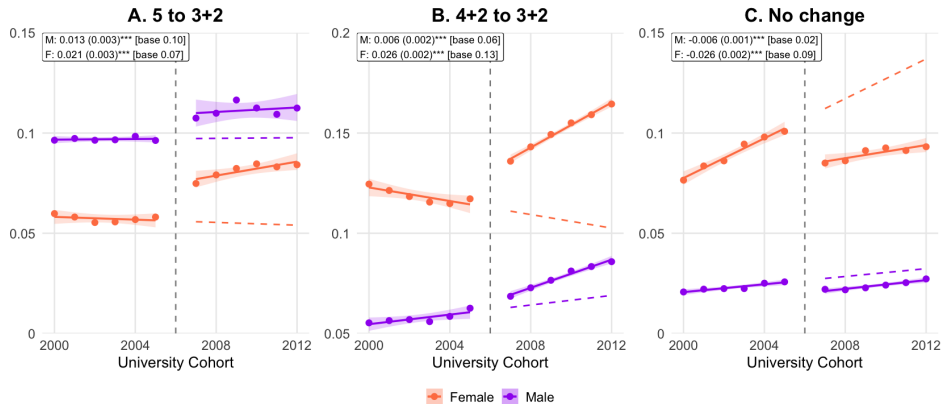
Gender ■ Female ■ Male

QP vs LFS vs IEFA

Likely low-SES

IEFA dropouts

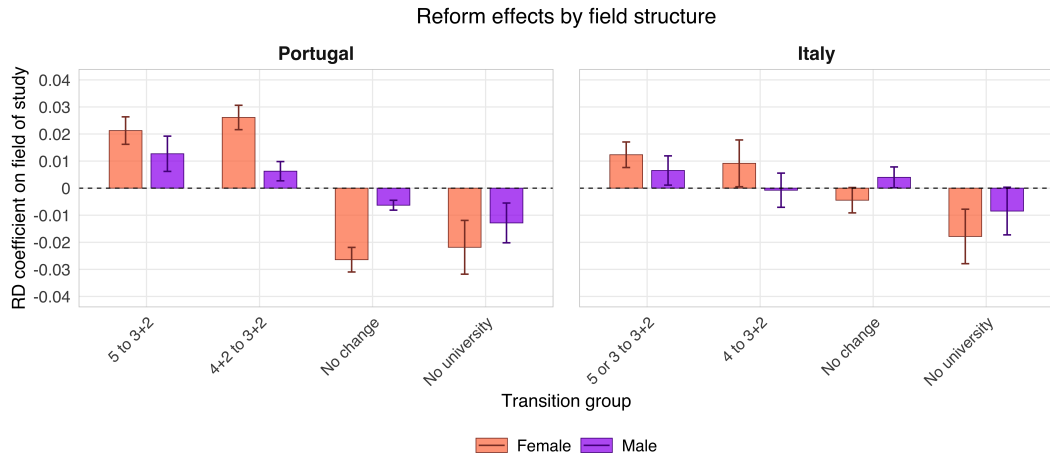
↑ reformed fields, ↓ non-reformed fields



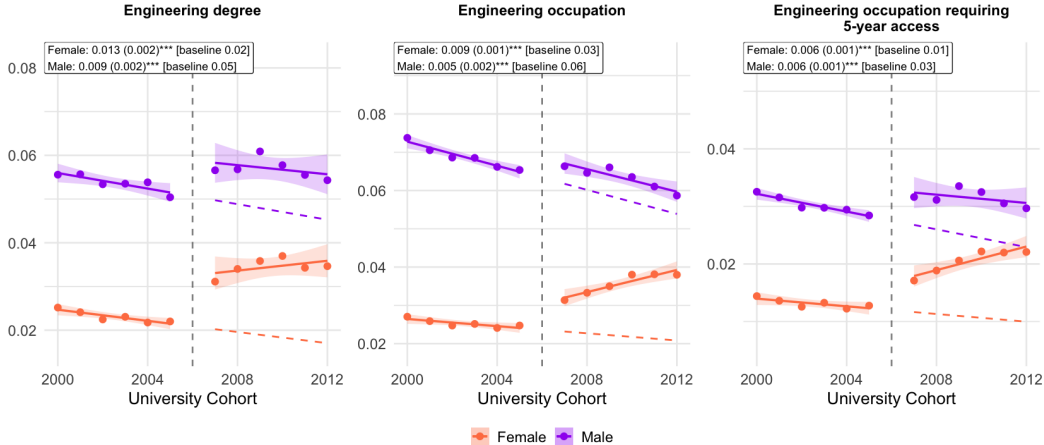
QP vs LFS vs IEFA

Detailed fields

External Validity: Portugal vs Italy

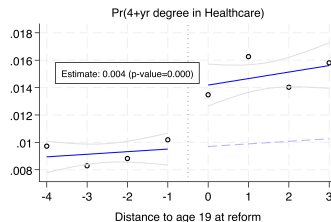
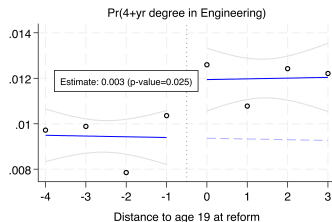
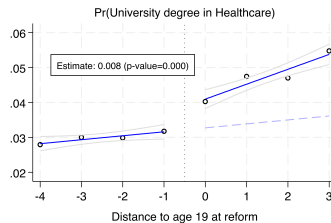
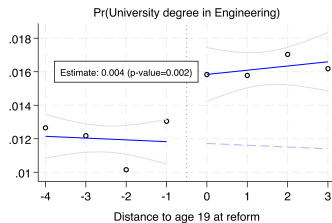


Students are not just stopping at the new 3-year degrees



Italy confirms this directly in degree length

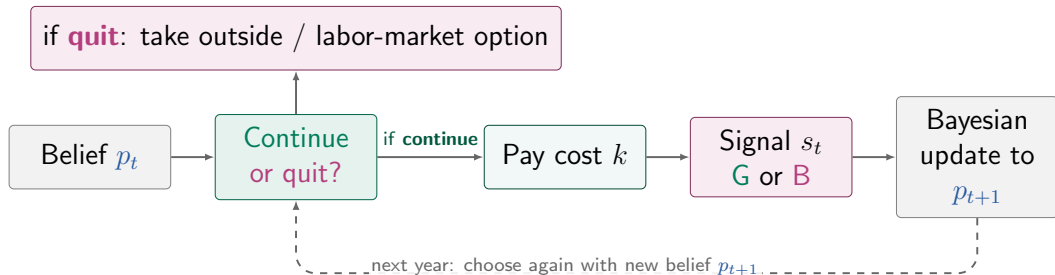
Female



A model of match uncertainty: Less to lose

Model overview: learning and stopping

University is a risky trial: students do not know their match quality, learn while enrolled, and decide each year whether continuing is still worthwhile.

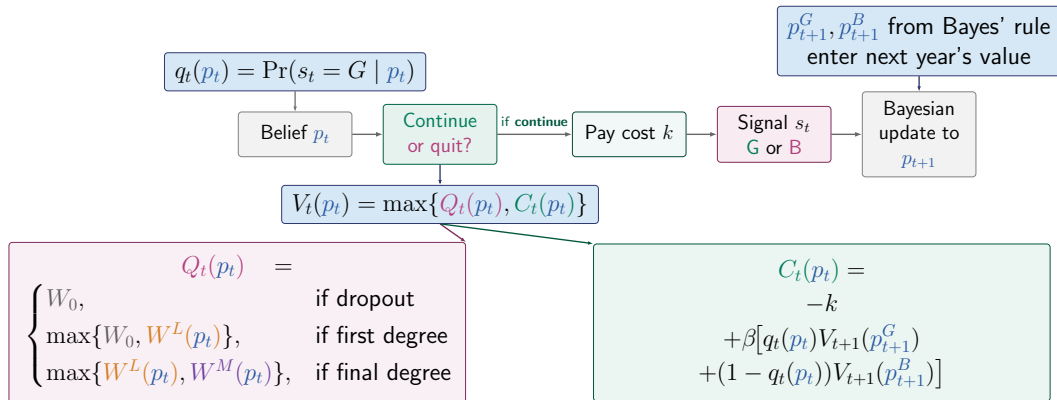


Priors

Learning equations

Model overview: learning and stopping

University is a risky trial: students do not know their match quality, learn while enrolled, and decide each year whether continuing is still worthwhile.



Priors

Learning equations

Why Bologna changes the enrollment bet

The key policy change is the timing of usable credentials.

Field	Before Bologna	After Bologna
STEM	Full credential after year 5	Lower credential at year 3; full at year 5
Non-STEM	Lower credential at year 4; full at year 6	Lower credential at year 3; full at year 5
Nursing	Full credential after year 4	Essentially unchanged

Bellman equation

Wage schedules

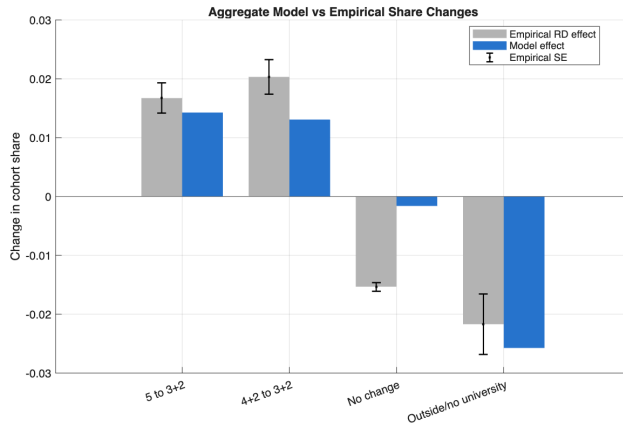
What disciplines the model

- ▶ **Calibrated (set externally):** discount factor, outside wage, wage payoff levels, prior beliefs, degree timing.
- ▶ **Solved to match exactly:** field intercepts → pre-reform shares.
- ▶ **Estimated by matching moments:** the field-choice scale, study costs, payoff slopes, and signal precision → the reform's RD effects on entry, field shares, years, and dropout.
- ▶ **Not targeted:** the wage RD — a check on the model.

Field choice

Estimation details

Model reproduces the reform's field shifts



Kappa robustness

Value gains

Moments

Model implies a +1.4 log-point wage RD

The wage RD is implied, not targeted.

$$\Delta \mathbb{E}[w] = \underbrace{\sum_{a \in \mathcal{A}} \Delta s_a \cdot \mathbb{E}[w \mid a, \text{pre}]}_{\substack{\text{share channel} \\ \text{more students/fields}}} + \underbrace{\sum_{a \in \mathcal{A}} s_a^{\text{post}} \cdot (\mathbb{E}[w \mid a, \text{post}] - \mathbb{E}[w \mid a, \text{pre}])}_{\substack{\text{composition channel} \\ \text{marginal entrants pull down averages}}}$$

Share: +1.9

+

Composition: -0.5

log points

=

Total: +1.4

$\mathcal{A}$ includes outside/no university and fields. Wage schedules are held fixed; the negative composition term comes from marginal entrants with lower predicted wage/match quality.

Full accounting table

Next step: when is stopping at 3 a mistake?

So far in the model, the reform is welfare-improving by construction. Adding an option cannot hurt students who optimize and update correctly.

Yet, the skeptical concern is: some students who would have completed 5 years before Bologna now stop at 3.

Next step: use the model to distinguish between efficient and mistaken early exits.

Efficient 3-year exit

Continuing to 5 was not worth it given the student's match quality or outside option.

Mistaken early exit

She would benefit from continuing, but stops at 3 because of, e.g.:

- ▶ unlucky bad signals
- ▶ impatience

Back to empirics

Wage decomposition & cumulative earnings

AKM: 30% of wage gains driven by sorting to better firms

Table: $\log w_{it} = g(\text{age}_{it}) + \alpha_i + \psi_{j(i,t)} + \mu_t + \epsilon_{it}$

	RD Jump		Share of wage effect (%)	
	Log Wage	Firm FE	Firm FE	Non-firm
Female sample				
RD Jump	0.021*** (0.001)	0.007*** (0.000)	31.6	68.4
Male sample				
RD Jump	0.013*** (0.001)	0.003*** (0.000)	26.4	73.6
p-val (F=M)	0.000	0.000		

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

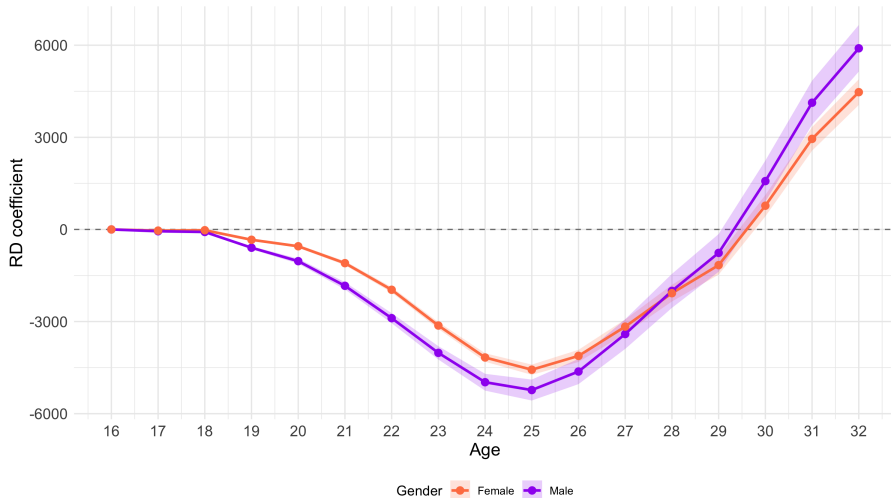
External validity: Italian female $\uparrow$ wages

	Pr(employed)	Log Wage	Avg yrs uni
Female			
RD Jump	0.002 (0.005)	0.021*** (0.007)	0.147*** (0.040)
Baseline	0.602	6.912	1.973
Male			
RD Jump	-0.003 (0.005)	0.001 (0.005)	0.041 (0.038)
Baseline	0.774	7.149	1.352
p-value Female = Male	0.555	0.022	0.055

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Earnings: ↓ in the 20s (longer in school), ↑ by early 30s

RD effects on cumulative earnings by age



Conclusion

- ▶ **Mechanism:** an intermediate exit gives students *less to lose* from enrolling.
- ▶ **Who enrolls:** +8–9% university completion, stronger for females.
- ▶ **What they study:** + reformed fields (STEM, Economics); – non-reformed (Healthcare).
- ▶ **Where they work:** sorting into better firms accounts for 30% of wage gains.
- ▶ **Cumulative earnings:** dip in the 20s, rise by early 30s.

Policy takeaway. Cutting degree length is not just about saving a year: a credible intermediate credential lowers the *downside risk* of enrolling.

Less to lose.

Appendix

Descriptives on selected sample [Back](#)

Variable	Control N = 8,074,453			Treatment N = 4,574,038		
	mean	min	max	mean	min	max
Age	30	17	41	27	17	34
Year	2014	2000	2023	2018	2006	2023
Female	0.48	0	1	0.47	0	1
University (any)	0.25	0	1	0.26	0	1
University (first-cycle)	0.20	0	1	0.18	0	1
Hourly wage	5.13	0.07	1036	5.93	0.13	1790
Age at first observation	22.66	17	41	22.19	17	41

No strategic delay: Pr(employed between HS and university) [Back](#)

	Probability
Whole sample	
RD Jump	0.001 (0.001)
Baseline	0.105
Female	
RD Jump	0.002 (0.002)
Baseline	0.128
Male	
RD Jump	-0.000 (0.002)
Baseline	0.085
p-value Female = Male	0.283

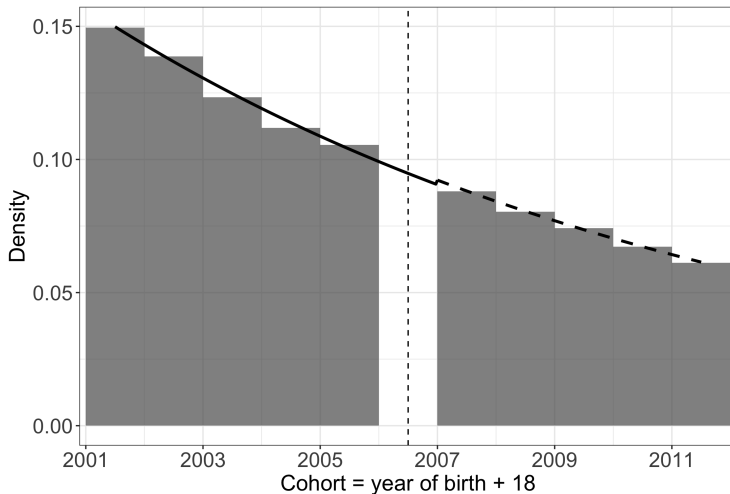
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

McCrary density test: no bunching at the cutoff

[Back](#)

McCrary density graph

Jump (log-density) = 0.019 (SE 0.020, $p = 0.379$)



	Probability that highest educ is:			Years of university:		
	Any university			Population average		
	QP	LFS	IEFA	QP	LFS	IEFA
Female: $N_{QP} = 6,153,392$; $N_{LFS} = 187,114$; $N_{IEFA} = 3,648$						
RD Jump	0.029*** (0.006)	0.037*** (0.004)	0.056 (0.041)	0.205*** (0.006)	0.185*** (0.024)	0.281 (0.263)
Baseline	0.340	0.229	0.287	1.919	1.938	1.811
Male: $N_{QP} = 6,797,302$; $N_{LFS} = 191,608$; $N_{IEFA} = 3,532$						
RD Jump	0.014*** (0.004)	0.012*** (0.003)	0.049 (0.053)	0.092*** (0.005)	0.070*** (0.020)	0.147 (0.357)
Baseline	0.200	0.133	0.121	1.274	1.608	1.730
p-val (F=M)	0.000	0.000	0.921	0.000	0.000	0.762

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

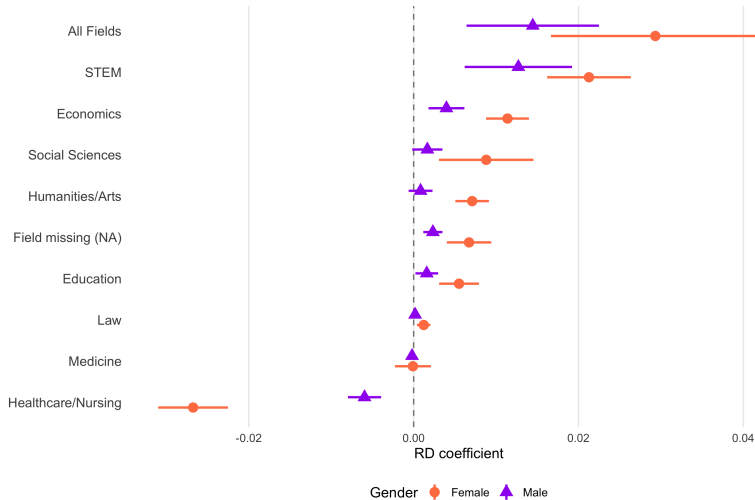
Field of Study: QP vs LFS vs IEFA [Back](#)

Probability that highest education is a university degree in:									
5 → 3+2: STEM			4+2 → 3+2: other			No change: health/law			
QP	LFS	IEFA	QP	LFS	IEFA	QP	LFS	IEFA	
Female: $N_{QP} = 6,011,539$; $N_{LFS} = 161,506$; $N_{IEFA} = 3,648$									
RD Jump	0.021*** (0.000)	0.008*** (0.002)	0.024 (0.021)	0.029*** (0.001)	0.037*** (0.004)	0.024 (0.037)	-0.027*** (0.001)	-0.010*** (0.003)	-0.001 (0.025)
Baseline	0.059	0.027	0.022	0.139	0.111	0.189	0.102	0.062	0.076
Male: $N_{QP} = 6,636,952$; $N_{LFS} = 167,591$; $N_{IEFA} = 3,532$									
RD Jump	0.012*** (0.001)	0.002 (0.002)	0.017 (0.014)	0.007*** (0.000)	0.009*** (0.002)	0.032 (0.035)	-0.006*** (0.000)	-0.003** (0.001)	0.002 (0.017)
Baseline	0.095	0.049	0.054	0.064	0.051	0.063	0.025	0.013	0.004
p-val (F=M)	0.000	0.077	0.802	0.000	0.000	0.872	0.000	0.012	0.908

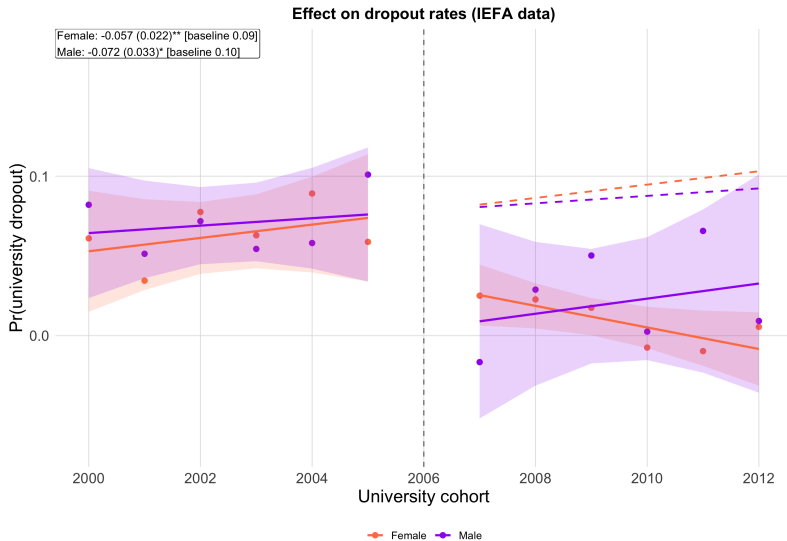
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

Bologna reform: RD coefficients by field of study

Points = RD estimate, bars = 95% CI



↓ dropouts (IEFA data) [Back](#)



Who are the marginal students? [Back](#)

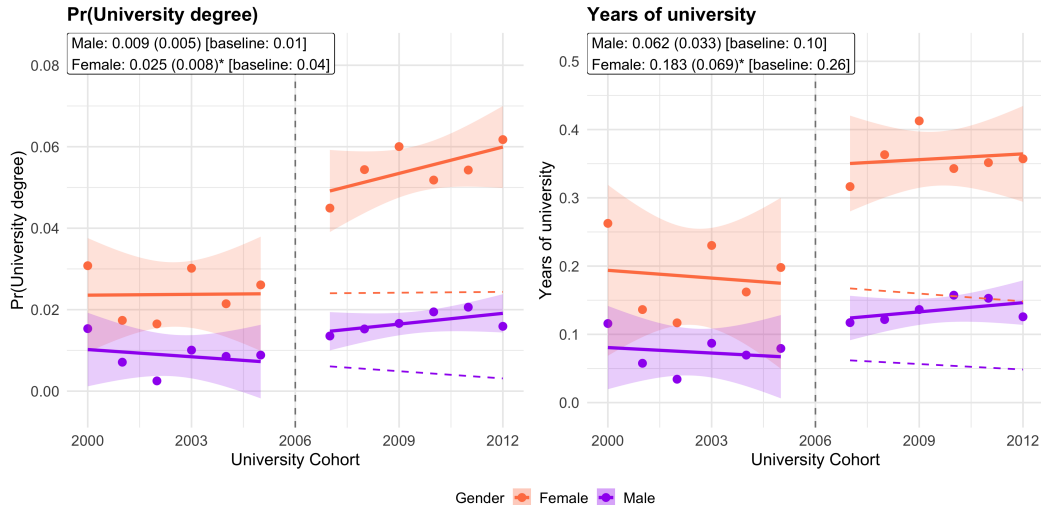
- ▶ **Low-SES proxy:** started working before age 18
 - ▷ Low-SES students should benefit more from time compression
- ▶ Reform does not change the share of Low-SES students:

	Effect of reform on: low-SES share
Female RD Jump	0.001 (0.009)
Female Baseline	0.070
Male RD Jump	-0.014 (0.012)
Male Baseline	0.104

Notes: Standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

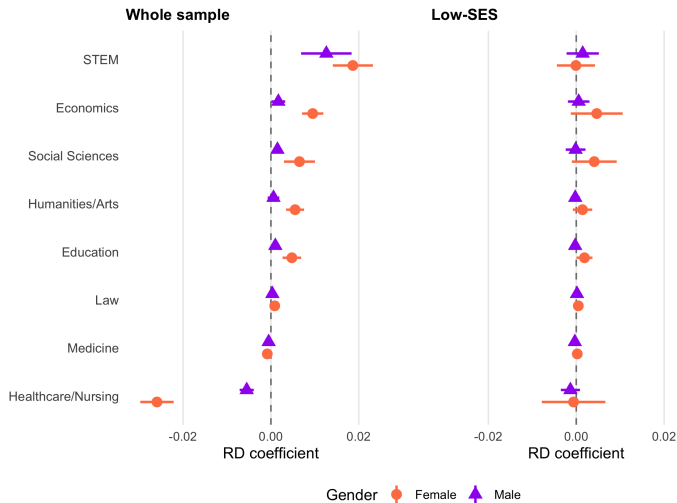
Low-SES: ↑ 62% university attainment for females [Back](#)



Low-SES: ↑ Economics and Social Sciences

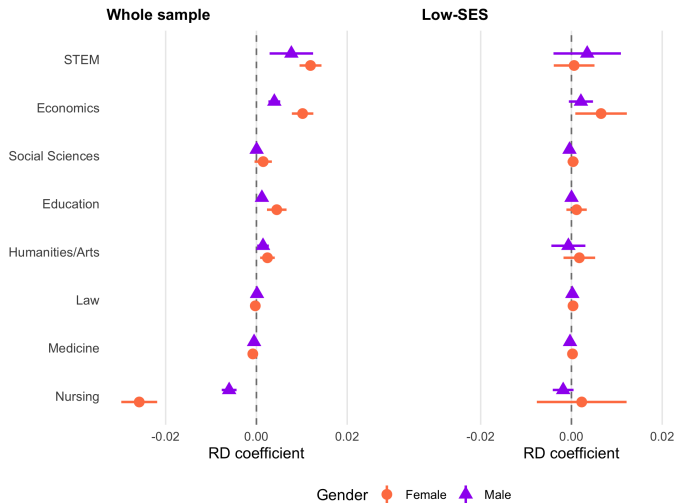
[Back](#)

RD coefficients: Probability of graduating from field X



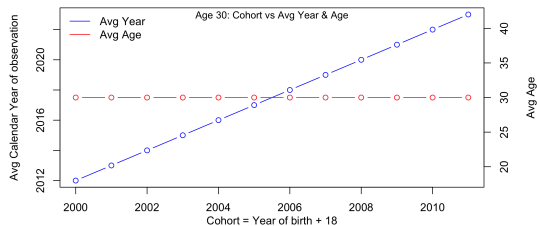
Field reallocation reflected in occupational upgrading [Back](#)

RD coefficients: Probability of working in occupation X



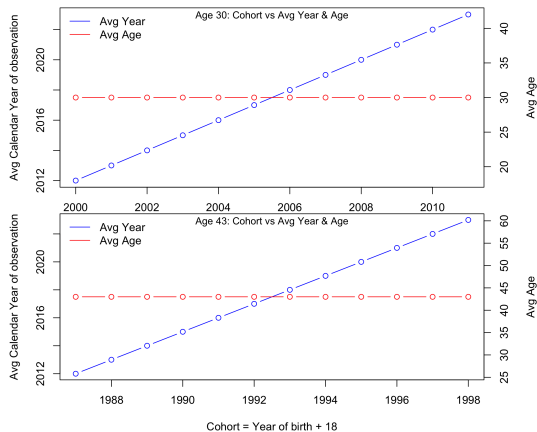
Wage variable

Cum Earn



Wage variable

Cum Earn



Relative cumulative earnings: same pattern

Cum Earn



AKM decomposition and age residualization [Back](#)

Step 1: remove the life-cycle wage profile.

$$\log w_{it} = g(\text{age}_{it}) + \tilde{w}_{it}, \quad \tilde{w}_{it} \equiv \log w_{it} - \hat{g}(\text{age}_{it}).$$

Step 2: decompose the same age-adjusted wage object.

$$\tilde{w}_{it} = \alpha_i + \psi_{j(i,t)} + \mu_t + \epsilon_{it}.$$

Step 3: run the RD on the object and on its components.

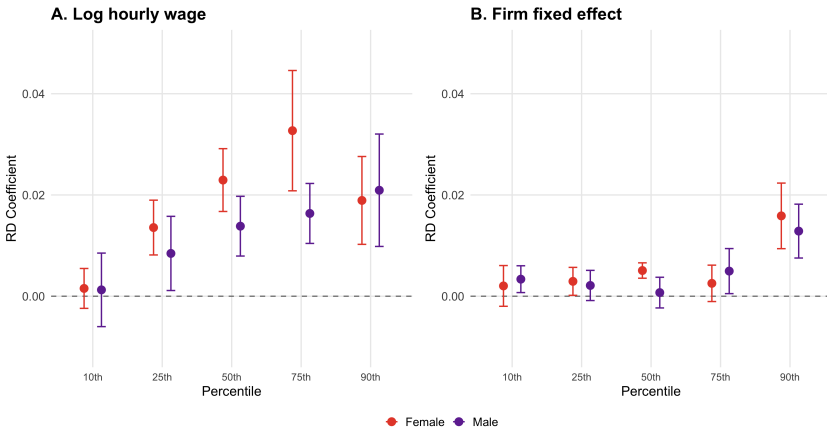
$$\Delta_{RD}^w = \underbrace{\Delta_{RD}^{\psi}}_{\text{firm component}} + \underbrace{\Delta_{RD}^{w-\psi}}_{\text{remaining non-firm component}}, \quad \text{Firm share} = \frac{\Delta_{RD}^{\psi}}{\Delta_{RD}^w}.$$

The component regressions do not add age controls because age has already been removed from the wage object being decomposed. The remaining component includes worker effects, time effects, and residual wage variation.

Females: wage gains strongest in the middle; firm gains at the top

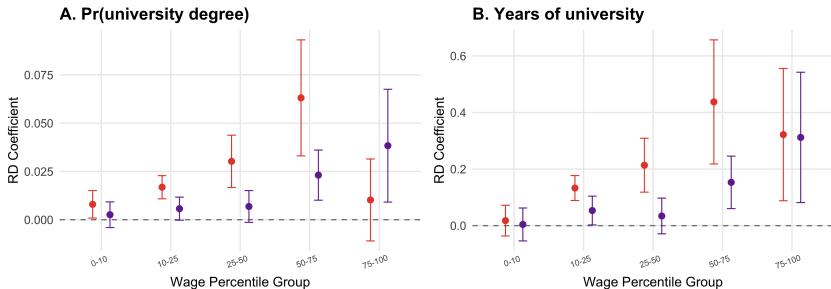
[Back](#)

Reform Effects on Wages and Firm Sorting by Percentile



Females: university gains concentrated in the middle [Back](#)

Reform Effects on Education by Wage Percentile Group

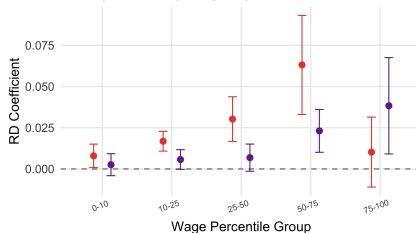


Notes: Percentile groups computed within cohort and gender. Bars show 95% CI.

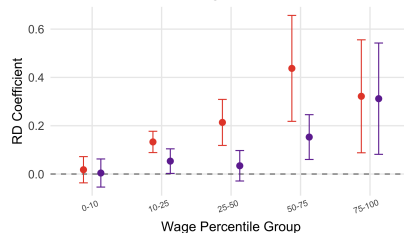
Females: ↑ STEM/Econ at top, ↓ Health at top [Back](#)

Reform Effects on Education and Field Composition by Wage Percentile Group

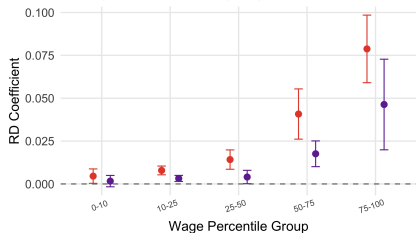
A. Pr(university degree)



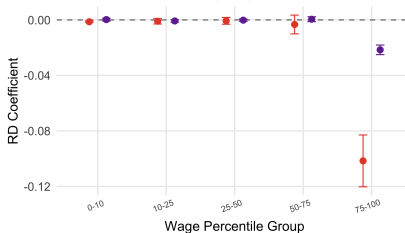
B. Years of university



C. Pr(STEM/Econ degree)



D. Pr(Healthcare degree)



Female Male

Notes: Percentile groups computed within cohort and gender. Bars show 95% CI.

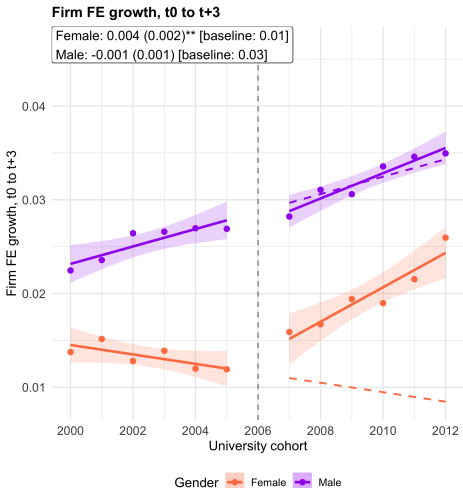
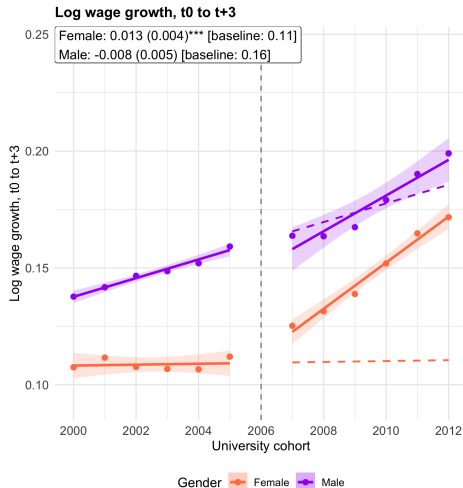
Early-career dynamics

Both genders: ↑ entry wages and better-quality firms

	Raw		Age-residualized	
	Log wage	Firm FE	Log wage	Firm FE
Female	0.028*** (0.002)	0.007*** (0.001)	0.005*** (0.002)	0.004*** (0.001)
Male	0.035*** (0.002)	0.006*** (0.001)	0.008*** (0.002)	0.002* (0.001)

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.
Age-residualized outcomes are residualized on age at first observation and its square.

Females: faster wage growth and faster climb up the firm ladder



Δ Wage stayers & Δ Wage movers

	Pr(changed firm) (by +3y)	Δ Log(hwage) Stayer	Δ Log(hwage) Mover
Female sample			
RD Jump	-0.002 (0.007)	0.020*** (0.005)	0.008* (0.004)
Male sample			
RD Jump	-0.016** (0.005)	-0.007 (0.006)	-0.007* (0.004)
p-val (F=M)	0.141	0.001	0.007

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

Δ computed as value at entry + 3 years minus value at entry.

AKM: decomposing wage growth

	RD Jump		Share of wage growth (%)	
	$\Delta \text{Log(hwage)}$	$\Delta \text{Firm FE}$	Firm FE	Non-Firm FE
Female sample				
RD Jump	0.013*** (0.004)	0.004** (0.002)	31.9	68.1
Male sample				
RD Jump	-0.008 (0.005)	-0.001 (0.001)	11.7	88.3
p-val (F=M)	0.000	0.013		

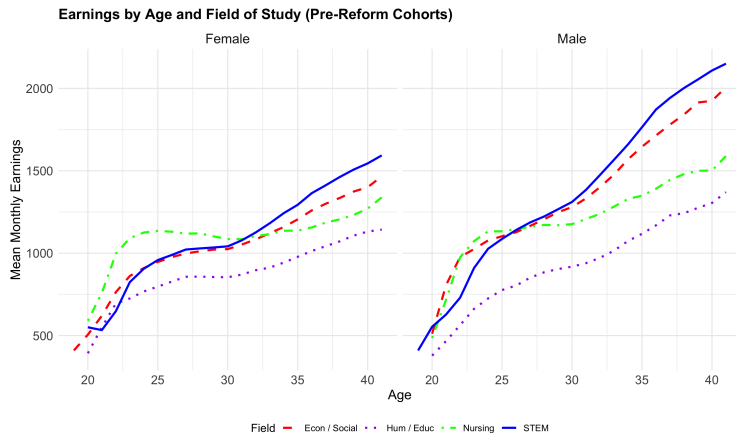
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors in parentheses.

Δ computed as value at entry + 3 years minus value at entry.

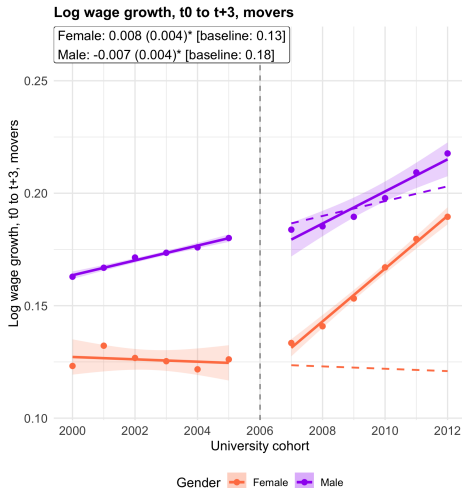
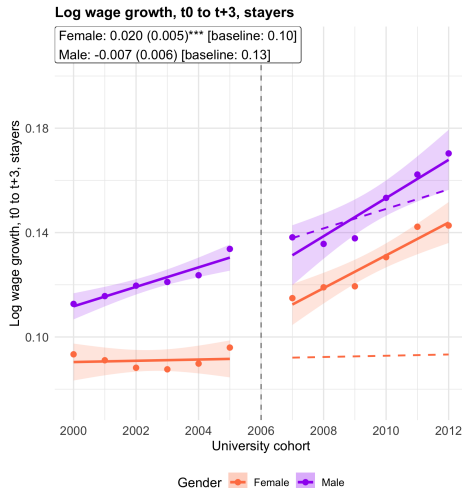
[Back to AKM decomposition](#)

[Back to graph](#)

Nursing: higher entry wages but slower long-run growth

[Back](#)

Females: wage growth stayers > movers



Types and prior beliefs Back

Latent type. Each student has a match type with the field she chooses:

$$\theta_f \in \{H, L\}, \quad p \equiv \Pr(\theta_f = H).$$

Priors. The student does not observe θ_f directly; she only knows her prior p_0 :

$$p_0 \sim \text{Beta}(\mu\kappa, (1 - \mu)\kappa).$$

- ▶ Mean: $\mu = \Pr(\text{expects to complete university}) = 0.39$ in PISA data.
- ▶ Dispersion: weighted logits for $p_i = \Pr(\text{uni} \mid X_i)$ using PISA test scores.
- ▶ $\text{sd}(\hat{p}_i) = 0.243$, so $\kappa \simeq 3.06$ and $p_0 \sim \text{Beta}(1.22, 1.83)$.

Signals and Bayesian updating [Back](#)

Signal technology. Each year of study delivers $s_t \in \{G, B\}$:

$$\Pr(s_t = G \mid H, f) = \lambda_{Hf}, \quad \Pr(s_t = G \mid L, f) = \lambda_{Lf}, \quad \lambda_{Hf} > \lambda_{Lf}.$$

Probability of next signal at belief p :

$$q_f(p) = p\lambda_{Hf} + (1 - p)\lambda_{Lf}.$$

Bayes update.

$$p_f^G = \frac{p\lambda_{Hf}}{q_f(p)}, \quad p_f^B = \frac{p(1 - \lambda_{Hf})}{1 - q_f(p)}.$$

- Signal precision $\lambda_{Hf} - \lambda_{Lf}$ governs how much beliefs move per signal.

The yearly choice: Bellman equation [Back](#)

Inside field f , regime r , the student chooses each t between quitting and continuing:

$$V_{f,t,r}(p) = \max \left\{ \underbrace{Q_{f,t,r}(p)}_{\text{quitting}}, \underbrace{C_{f,t,r}(p)}_{\text{continuing}} \right\}.$$

$$C_{f,t,r}(p) = -k_f + \beta \left[q_f(p) V_{f,t+1,r}(p_f^G) + (1 - q_f(p)) V_{f,t+1,r}(p_f^B) \right].$$

$$Q_{f,t,r}(p) = \begin{cases} W_0, & t < L_{f,r} \text{ (no credential yet)} \\ \max\{W_0, W_f^L(p)\}, & L_{f,r} \leq t < T_{f,r} \\ \max\{W_f^L(p), W_f^M(p)\}, & t = T_{f,r} \text{ (graduation)} \end{cases}$$

From within-field values to field shares [Back](#)

At entry, a student with belief p_0 gets

$$U_{if,r}(p_0) = V_{f,0,r}(p_0) + a_f + \varepsilon_{if}, \quad U_{i0} = W_0 + \varepsilon_{i0},$$

where ε are iid Type-I extreme value with scale σ . Hence:

$$\Pr(f \mid p_0, r) = \frac{\exp((V_{f,0,r}(p_0) + a_f)/\sigma)}{\exp(W_0/\sigma) + \sum_j \exp((V_{j,0,r}(p_0) + a_j)/\sigma)}.$$

Aggregate field shares integrate over the prior distribution:

$$P_{f,r}^{agg} = \int \Pr(f \mid p_0, r) dF(p_0).$$

Field intercepts a_f are solved so that pre-reform aggregate shares are matched exactly.

Model object	How set	Empirical discipline	Outer loss?
β, W_0	fixed	discounting; outside wage normalization	No
p_0 distribution	fixed per run	PISA: $\mu = 0.399, \kappa \simeq 3.06$	No
Degree menus (L, T)	fixed	Bologna credential timing by field	No
Field intercepts a_f	inner solve	exact match to 2005 pre-reform shares	No residual
Payoff intercepts α_f^M, α_f^L	mechanical	match field wage ratios at $\bar{p}_0$	No
σ	free	substitution scale in entry choice	Yes
Costs k_f	3 free parameters	dropout and entry response	Yes
Full payoff slopes γ_f^M	3 free parameters	field-specific RD responses	Yes
Lower slopes γ_f^L	restricted	$\gamma_f^L = \rho_f \gamma_f^M$	No
Signal spreads $\lambda_{Hf} - \lambda_{Lf}$	3 free parameters	learning, dropout, field reallocations	Yes

Outer loss matches: 4 RD share effects, 3 RD share $\times$ years effects, total entry, pre-dropout, and dropout change.

Identification

What identifies the behavioral parameters?

[Back](#)

Parameter block	Economic role	Primarily disciplined by
σ	Substitution across fields and outside option	Total entry and relative field-share responses
$\{k_f\}$	Field-specific cost of continuing	Entry response, pre-reform dropout, and dropout change
$\{\gamma_f^M\}$	Return to being a high-match type in field f	Field-specific RD responses
$\{\lambda_f^H - \lambda_f^L\}$	How quickly students learn match quality	Dropout and share-by-years responses

$$\hat{\theta} = \arg \min_{\theta} \left[m^{model}(\theta) - m^{data} \right]' W \left[m^{model}(\theta) - m^{data} \right].$$

Note: “primarily disciplined by” $\neq$ one-to-one identification. All free parameters affect multiple moments.

Model wage RD accounting [Back](#)

$$\Delta \mathbb{E}[w] = \underbrace{\sum_f \Delta \text{share}_f \cdot \mathbb{E}[w \mid f, \text{pre}]}_{\text{share channel}} + \underbrace{\sum_f \text{share}_f^{\text{post}} \cdot (\mathbb{E}[w \mid f, \text{post}] - \mathbb{E}[w \mid f, \text{pre}])}_{\text{composition channel}}$$

Channel	$\Delta \log w$ (log points)
Share channel (between)	+1.9
Composition channel (marginal entrants)	-0.5
Model wage RD (field-channel total)	+1.4

Counterfactual use: candidate mistaken exits [Back](#)

Use the fitted model as a laboratory: simulate the same student under alternative information or preference environments, then compare exit decisions.

Mechanism	Counterfactual benchmark	Object to report
Inefficient learning	Correct Bayesian updating using the true signal process	Exits induced by distorted beliefs
Bad shocks for high types	Same optimal rule, but reveal true latent match type	False-negative dropout among early exits
Impatience / regret	Long-run preferences, e.g. $\beta = 0.95$	Continuation value lost by impatient exits

[Inefficient learning](#)[High-type bad shocks](#)[Impatience / regret](#)

Mistaken exits I: inefficient learning [Back](#)

Students choose using **perceived** beliefs, while true match quality and payoffs are governed by the true signal process.

Baseline Bayesian benchmark:

$$\text{logit}(p^G) = \text{logit}(p) + \log\left(\frac{\lambda_{Hf}}{\lambda_{Lf}}\right), \quad \text{logit}(p^B) = \text{logit}(p) + \log\left(\frac{1 - \lambda_{Hf}}{1 - \lambda_{Lf}}\right).$$

Distorted learning example: overweight bad signals

$$\text{logit}(\tilde{p}^B) = \text{logit}(\tilde{p}) + \omega_B \log\left(\frac{1 - \lambda_{Hf}}{1 - \lambda_{Lf}}\right), \quad \omega_B > 1.$$

$$\text{or more generally: } \tilde{\lambda}_{Hf} - \tilde{\lambda}_{Lf} \neq \lambda_{Hf} - \lambda_{Lf}.$$

Define an inefficient-learning exit when the student exits under $\tilde{p}_t$, but would continue under the correctly updated p_t .

$$1\{\tilde{C}_{f,t}(\tilde{p}_t) < \tilde{Q}_{f,t}(\tilde{p}_t)\} \cdot 1\{C_{f,t}(p_t) \geq Q_{f,t}(p_t)\}.$$

Mistaken exits II: bad shocks for high types [Back](#)

This is the cleanest object in the current model. True match quality exists, but students only observe stochastic signals.

- ▶ A high type, $\theta_i = H$, can receive a bad signal history such as B, B .
- ▶ Given those signals, exit can be fully rational under correct Bayesian updating.
- ▶ Ex post, the exit is a **false negative**: the model knows the student was actually high match.

False-negative dropout share:

$$\Pr(\theta_i = H \mid \text{early exit}) = \frac{\sum_i 1\{\theta_i = H\} 1\{\text{exit before credential}\}}{\sum_i 1\{\text{exit before credential}\}}.$$

Bologna diagnostic:

$$\Pr(\theta_i = H, \text{no-credential exit}^{pre}, \text{credential or continue}^{post}).$$

This measures high-match students whose unlucky early histories are partly insured by the shorter path to a credential.

Mistaken exits III: impatience / regret [Back](#)

Allow students to differ in patience or present bias. The same student may exit under short-run preferences but continue under a long-run self.

Impatient choice rule:

$$\text{exit if } C_{f,t}^{short}(p_t; \beta_i) < Q_{f,t}(p_t), \quad \beta_i < \beta^{LR}.$$

Long-run benchmark: evaluate the same state using $\beta^{LR} = 0.95$:

$$R_{i,t} = C_{f,t}^{LR}(p_t; \beta^{LR}) - Q_{f,t}(p_t).$$

Regretted exit:

$$1\{\text{exit under } \beta_i\} \cdot 1\{R_{i,t} > 0\}.$$

Report the share of exits with $R_{i,t} > 0$, the mean positive regret gap, and whether Bologna reduces regret by making continued study less costly before the first credential.

Credential payoffs linear in graduation belief:

$$W_f^M(p) = \alpha_f^M + \gamma_f^M p, \quad W_f^L(p) = \alpha_f^L + \gamma_f^L p, \quad W_0 = 1.$$

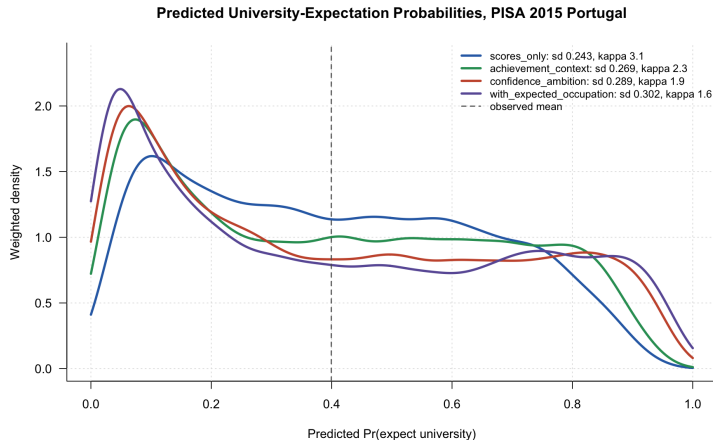
Pinned by wage data. Intercepts chosen so that at $\bar{p}_0 = 0.6$,

$$W_f^M(\bar{p}_0) = \frac{\overline{\text{wage}}_f^M}{\overline{\text{wage}}_{\text{Outside}}}, \quad W_f^L(\bar{p}_0) = \frac{\overline{\text{wage}}_f^L}{\overline{\text{wage}}_{\text{Outside}}}.$$

Slopes γ_f^M free; $\gamma_f^L = \rho_f \gamma_f^M$ with $\rho_{\text{STEM}} = 0.6$, $\rho_{\text{non-STEM}} = 0.5$.

Key restriction: wage *schedules* are the same pre and post. Average observed wages can still shift through composition (partial equilibrium).

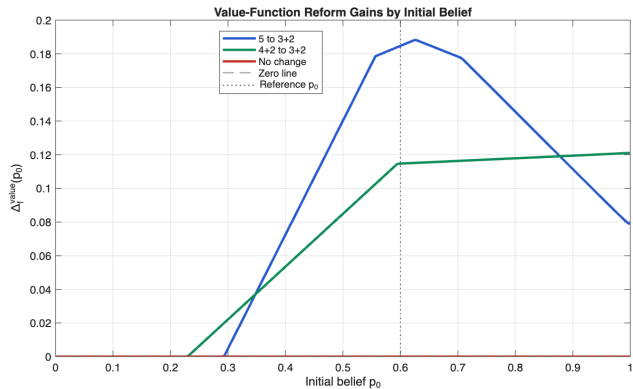
Prior beliefs using PISA data [Back](#)



- Variables per specifications: (i) test scores only, (ii) + SES and grade repetition, (iii) + ambition and confidence index, (iv) + expected occupation
- $\kappa \in [1.63, 3.06]$.

κ	Prior sd	Δ univ. entry
1.63	0.302	+1.97 pp
1.87	0.289	+1.92 pp
2.32	0.269	+2.04 pp
3.06	0.243	+2.13 pp

Reform gains by initial belief [Back](#)



- Avdeev, S. (2025). Balancing study and work: Heterogeneous impact of the bologna reform on the labour market. *Oxford Bulletin of Economics and Statistics*, 87(1):252–286.
- Bondonio, D. and Berton, F. (2018). The impact of degree duration on higher education participation: Evidence from a large-scale natural experiment. *Oxford Bulletin of Economics and Statistics*, 80(5):905–930.
- Cappellari, L. and Lucifora, C. (2009). The “bologna process” and college enrollment decisions. *Labour economics*, 16(6):638–647.
- Cardoso, A. R., Portela, M., Sá, C., and Alexandre, F. (2008). Demand for higher education programs: the impact of the bologna process. *CESifo Economic Studies*, 54(2):229–247.
- Jepsen, C., Troske, K., and Coomes, P. (2014). The labor-market returns to community college degrees, diplomas, and certificates. *Journal of Labor Economics*, 32(1):95–121.
- Manski, C. F. (1989). Schooling as experimentation: a reappraisal of the

postsecondary dropout phenomenon. *Economics of Education review*, 8(4):305–312.

Mountjoy, J. (2022). Community colleges and upward mobility. *American Economic Review*, 112(8):2580–2630.

Stevens, A. H., Kurlaender, M., and Grosz, M. (2019). Career technical education and labor market outcomes: Evidence from california community colleges. *Journal of Human Resources*, 54(4):986–1036.

Stinebrickner, T. and Stinebrickner, R. (2012). Learning about academic ability and the college dropout decision. *Journal of Labor Economics*, 30(4):707–748.

Trachter, N. (2015). Stepping stone and option value in a model of postsecondary education. *Quantitative Economics*, 6(1):223–256.